Throughput Is Not Liveness:
Three Clocks for GPU-Free Music Generation on iPhone

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Abstract

Real-time generative audio is usually reduced to one question: can inference run faster than playback? That criterion is necessary and insufficient. A system may average faster than real time while buffering tens of seconds of stale audio, and it may deliver every sample while a state-boundary error changes the sound. We present an end-to-end deployment and falsification study of the 230-million-parameter `mrt2_small` live-music model on iPhone. The system combines a 12-layer temporal transformer, a 12-level residual-vector-quantized depth rollout, SpectroStream decoding, inverse STFT, and 48 kHz stereo delivery. A pure fixed-shape temporal graph executes on the Apple Neural Engine (ANE), while Swift owns its 41-frame K/V ring. Ten of ten cold processes admit the graph on both A17 Pro and A14, and a model-attributed trace contains temporal and decoder ANE work with no application-attributed GPU interval.

On an iPhone 15 Pro Max, the corrected runtime sustains 610 seconds in the foreground, reaches the `serious` thermal state, completes 628 nominal one-second producer iterations in 609.65 seconds, and records zero underruns or drops. The p99 of iteration-normalized active stage cost is 24.81 ms per token against a 40 ms throughput budget. We emphasize what this number is not: it is the p99 of 25-token iteration means, not per-token tail latency. Backpressure bounds the queue near 21 seconds, so the experiment proves continuous throughput, not low-latency steering. Separate device evidence measures controller application in 4-15 ms but audible changes only after 6.48-9.02 seconds without queue discard.

The main scientific result comes from a reviewer-motivated 2x2 crossover. For three seeds, we generate 600-second token trajectories with MLX and the shipped Core ML port, then decode each through stateful MLX or the stateless phone boundary. Fixed-token graph and DSP controls separate sampling, numerical, windowing, and reconstruction hypotheses. The pulse excess follows stateless decoder windowing in all 60 paired 30-second windows: median seed effect +0.01706, compared with +0.00018 for the Core ML graph itself. A direct tensor probe shows that the original 25-token calls discard causal convolution history; retaining 12 token frames raises correlation with stateful MLX from 0.108 to above 0.999999999. Adding that context reduces pulse share by 0.01650 and changes the original diagnostic-limit count from 35/60 windows to 13/60, versus 11/60 for stateful MLX. A new 600-second iPhone capture matches the stateful reference count at 5/20 windows for the principal seed, with no dropout. Thus a result first framed as long-horizon model degeneration was a systems boundary bug. The broader lesson is methodological: live generation has separate compute, delivery, and generative clocks, and a credible study must localize failures across all three.

1 Introduction

Live music generation is an unusually unforgiving systems workload. Its autoregressive model must keep producing, its audio callback must never wait, steering should become audible while the intent is still relevant, and the trajectory must remain musically plausible. A batch generator can finish late. A live generator that finishes late clicks or falls silent; one that buffers too aggressively plays an obsolete decision; one with incorrect state can run perfectly on time while sounding wrong.

Magenta RealTime introduced continuously steerable causal music generation [Lyria Team et al., 2025]. Magenta RealTime 2 (MRT2) provides open weights and an MLX implementation for Apple Silicon Macs [Google Magenta, 2026]. Its small configuration emits 12 residual-vector-quantized (RVQ) codes at 25 Hz, and a SpectroStream decoder reconstructs 48 kHz stereo audio [Google, 2026]. This paper asks whether the complete loop can execute continuously on an iPhone without application GPU work, and, more importantly, what evidence is needed to make that statement mean anything.

Our first answer was wrong in an instructive way. A signed A17 Pro runtime sustained 610 seconds with zero underruns and a growing reservoir. Its captured audio nevertheless crossed a predeclared 4-16 Hz envelope-pulse threshold and failed three of five exploratory automated listening votes. We reported a failed "generative clock." That framing survived many engineering controls but not the decisive one: we had never run the reference sampler to the same horizon and crossed token source with decoder implementation.

We therefore freeze three causal hypotheses. H1 is model-intrinsic or sampling-induced recurrence: the pulse should follow tokens into either decoder. H2 is numerical divergence in the temporal/depth port: the Core ML token source should be worse after decoding with clean MLX. H3 is a decoder or DSP boundary error: the pulse should follow the stateless phone decoding path for either fixed token source. We then split H3 into an MLX FLOAT32 pre-iSTFT graph, the Core ML FP16 graph, the production C++ reconstruction core, and a measured context intervention. The result strongly supports a refined H3: a stateless causal decoder was invoked with insufficient left context.

This correction changes the paper's thesis. "Faster than playback" is not a complete real-time claim, and a failed audio detector is not a causal label. We define three clocks:

1. **Compute clock.** Sustained production rate must be at least playback rate. Batch-normalized stage costs describe headroom but do not become per-token tail latency by division.
2. **Delivery clock.** The callback must observe zero underruns and producer drops, while queued audio remains inside a band whose upper bound is chosen from an audible steering-latency budget.
3. **Generative clock.** Prompt-conditional audio measures and controlled comparisons must remain valid across the horizon. A detector can identify a symptom; only an intervention or crossover can identify its source.

We make four contributions:

- A complete GPU-free placement and sustained-throughput study of MRT2 on A17 Pro, with a pure temporal graph, host-owned K/V state, in-process admission gates, and model-attributed accelerator evidence.
- A duration-controlled thermal result: an A17 CPU+GPU-policy control has 38.51 ms p99 iteration-normalized cost in five short processes, but 49.23 ms over a matched 610-second run and crosses 40 ms only after minute six.
- A three-seed, long-horizon token-by-decoder crossover that localizes a plausible generative failure to missing causal decoder history, plus a 12-frame overlap intervention verified at tensor, audio, and physical-device levels.
- A corrected liveness contract that distinguishes continuous playback from interactive steering. The current runtime sustains output, but its ordinary 6.48-9.02 second audible control latency is not yet a strong live-product result.

The weights are unchanged. Previous conversion and accelerator-admission findings appear in the companion *Surgical Inference* manuscript [Mireles, 2026]. Here the units of contribution are the composed runtime, sustained physical-device evidence, causal audio localization, and measurement contract.

$$\mathcal{C}_{\text{compute}} \equiv R_{\text{producer}} \geq 1, \tag{1}$$

$$\mathcal{C}_{\text{delivery}} \equiv (U = 0) \wedge (D = 0) \wedge (0 < B(t) \leq B_{\max}), \tag{2}$$

$$\mathcal{C}_{\text{gen}} \equiv \bigwedge_{k=1}^{K} m_k \in \mathcal{A}_k. \tag{3}$$

Here R_{producer} is sustained generated-audio rate relative to playback, U and D are callback underruns and producer drops, B is queued PCM, and $B_{\max}$ is chosen from an audible steering-latency budget. Each m_k is a prompt-conditional audio measure with acceptance set $\mathcal{A}_k$. No score from one clock substitutes for another.

2 Background and related work

2.1 Live music and neural audio

MusicGen generates text- or melody-conditioned music from interleaved codec tokens [Copet et al., 2023]. RAVE demonstrated 48 kHz neural synthesis faster than real time on a laptop CPU [Caillon and Esling, 2021]. Live Music Models instead make causality and continuous steering part of the product contract [Lyria Team et al., 2025]. MRT2 is the open model used here. Its SpectroStream codec operates in the time-frequency domain and supports full-band stereo [Li et al., 2025], extending the streamable neural-codec lineage of SoundStream and EnCodec [Zeghidour et al., 2022] [Défossez et al., 2023].

Autoregressive generation can amplify model or decoding errors over a long horizon [Rohatgi et al., 2025]. Repetitive text degeneration motivated dynamic nucleus sampling [Holtzman et al., 2020]. These results make recurrence a credible hypothesis for a periodic audio symptom, not a conclusion. In our case, both MLX and Core ML trajectories retain full distinct-frame ratio, about 6.2-6.4 bits of mean per-level entropy in late complete windows, and no exact period-1 through period-8 cycles. More decisively, the symptom follows decoder windowing under fixed tokens.

2.2 Mobile heterogeneous inference

MLPerf Mobile emphasizes that on-device performance is a property of the full software-hardware stack [Reddi et al., 2022]. Apple documents fixed-shape transformer deployment patterns for the ANE [Apple Machine Learning Research, 2022, Apple, 2026]. Similar heterogeneous scheduling appears in mobile NPU work such as `llm.npu` [Xu et al., 2025]. MRT2 differs from mobile LLM prefill: it repeats a small causal step indefinitely, feeds an audio callback, and exposes every state or timing error to a listener. Requested compute units, graph conversion, and a short latency benchmark are therefore weak proxies. We separately measure admission, accelerator activity, sustained capacity, callback continuity, queue depth, and audio behavior.

3 System design

3.1 Workload and host boundary

At 25 Hz, one MRT2 token frame represents 1,920 output samples and 40 ms of 48 kHz stereo audio. The temporal transformer has 12 layers and consumes the previous frame's mean token embedding, source conditioning, and a local attention history. The depth transformer autoregressively selects 12 RVQ codes. Their 256-dimensional codebook rows are summed and passed to the SpectroStream decoder. fig. 1 shows the deployed boundary.

The temporal graph is a pure single-frame tensor function. Inputs contain the current temporal and source vectors, a fixed-shape validity bias, and 48 ordinary K/V arrays for self- and cross-attention across 12 layers. Outputs contain one temporal vector and 48 one-token cache updates. Swift owns a preallocated 41-frame chronological ring. This removes in-graph state mutation, which had compiled in a probe but failed ANE admission in the application.

The state proof asks two independent questions. First, identical input on fresh and warmed state must diverge, preventing a write-only-state false pass. Second, 64 consecutive predictions must match a frozen reference, including 23 after the ring wraps. Correlation is 0.9995 on both phones, and all outputs are finite. The runtime repeats this proof and queries the temporal compute plan at session start.

3.2 One-call depth rollout

Table 1: Measured A14 depth rollout ablation. The in-graph boundary removes host boundaries but still traverses transformer weights at each dependent RVQ level; the major gain is FLOAT16, not a fictitious single weight traversal.

Depth boundary	Precision	ms/token frame
12 separate full-pass predictions	FLOAT32	40.2
One in-graph 12-level prediction	FLOAT32	∼37.0
One in-graph 12-level prediction	FLOAT16	12.7

The depth transformer has dependent RVQ levels: level `k+1` consumes the code selected at level `k`. Twelve separate Core ML predictions on A14 cost 1,004 ms per 25-frame producer iteration, or 40.2 ms per token frame. A one-prediction graph unrolls the dependency, takes host-generated Gumbel noise, applies the fixed top-40 mask, gathers embeddings, and returns all 12 codes plus temporal feedback.

The measured ablation corrects an easy but false bandwidth story. The in-graph rollout does not turn 12 dependent transformer applications into one weight traversal. FLOAT32 still costs about 37 ms per frame. The large gain comes from the validated FLOAT16 artifact, which measures 12.7 ms on A14 and 8.4 ms on A17 Pro (table 1). FLOAT32 is token-exact in 900 comparisons; FLOAT16 flips near-tie tokens and was accepted only after distributional and device audio gates.

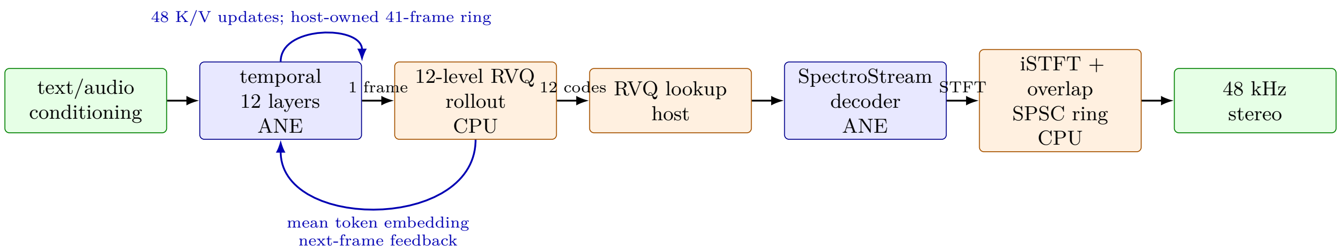

Figure 1: End-to-end deployment boundary. Blue stages are observed on the ANE; orange stages remain deliberate host/CPU work. Temporal recurrence crosses the model boundary as ordinary K/V tensors; the stateless decoder retains 12 token frames of measured causal context. No claim of an "all-ANE" system is made.

3.3 Decoder context and reconstruction

The original phone decoder accepted 25 token embeddings and emitted 96 STFT frames. The host advanced by 24 token frames, preserving only the one-frame SpectroStream lookahead. This treated the decoder as a finite-window function. Its convolutional stack is causal but stateful: output near a boundary depends on more history than the formal lookahead.

The corrected boundary retains 12 token frames. The first 25-token prediction initializes reconstruction state and emits its full 96 STFT frames. Each later prediction advances by 12 tokens, drops the first 48 context STFT frames, and emits the 48 new frames. The Core ML input shape and model artifact remain unchanged. A preallocated FLOAT32 crop buffer materializes logical output using the declared `MLMultiArray` shape and strides; allocation and DSP remain off the audio callback.

The C++ inverse STFT uses the periodic Hann synthesis window used by upstream SpectroStream and performs overlap-add into a lock-free single-producer, single-consumer PCM ring. During diagnosis we found a second parity debt: the old C++ core applied a normalized dual-window convention. Direct sample tests now match the upstream periodic-Hann reconstruction. However, this change alone *increases* pulse share by a median 0.00321 across seeds. It is a real parity fix but not the causal fix for the reported symptom.

3.4 Buffering and steering

Inference and reconstruction run on a producer task. The Core Audio callback only reads a bounded ring and zero-fills an underflow. A startup reservoir absorbs jitter. Once the ring reaches its high watermark, producer backpressure prevents unbounded memory growth.

That architecture guarantees neither low latency nor liveness. A high watermark of roughly 21 seconds allows 21 seconds of already-rendered intent. In an earlier two-event control test, the controller applied new state after 0.015 and 0.004 seconds, but matched-reference audible changes arrived after 6.479 and 9.023 seconds. A proof-mode queue discard retained 19,200 frames (0.4 seconds), faded in over 3,840 frames, discarded 647,936 frames, and ended with zero underruns and drops. The capture point was before the playback ring, so that receipt cannot prove click-free speaker output. We therefore report continuous buffered playback as solved and subsecond audible steering as open.

3.5 Ten-second temporal refresh

All principal device and crossover trajectories reset temporal K/V and the previous-frame feedback vector every 10 seconds. Decoder context, overlap-add, and queued PCM remain continuous. This condition must be stated because the 41-frame ring covers only 1.64 seconds. After 10 seconds, attention slots have already rolled over several times; the intervention's distinctive long-horizon effect is resetting the recurrent feedback loop and returning the cache to a cold start. It is a deployed bounded-trajectory protocol here, not evidence that an unrefreshed model is stable.

4 Experimental method

4.1 Devices, artifacts, and placement

The primary device is a 2023 iPhone 15 Pro Max (`iPhone16,2`, A17 Pro). The boundary device is a 2020 iPhone 12 Pro (`iPhone13,3`, A14). Signed release builds run in the foreground with the screen on. The fixed condition is `warm ambient texture`, certified `warm.bin` source conditioning, top-k 40, style guidance 3.1, temperature 1.0, and seed 20260718 unless a replication seed is named.

Placement has two receipts. Ten cold application processes query `MLComputePlan`; all ten assign temporal estimated cost 1.0 to the ANE on each phone and pass the state proof. A process-targeted Instruments trace then records temporal and decoder ANE predictions, Core ML activity for every model family, and an empty application Metal GPU table. The claim is GPU-free, not all-ANE: depth, Gumbel generation, RVQ lookup, cache mutation, inverse STFT, ring management, logging, and UI work remain on the CPU.

4.2 Throughput and the batch-normalized cost

One producer iteration generates 25 temporal/depth frames and one to three decoder calls. We preserve the historical diagnostic

$$t_{\text{eff}} = \frac{t_{\text{temporal}} + t_{\text{depth}} + t_{\text{sampling}} + t_{\text{decoder}}}{25}.$$

We rename it **iteration-normalized compute cost**. Its p99 is the p99 of a mean over each 25-token iteration. It suppresses within-iteration dispersion and cannot certify a per-token deadline. The actual compute-clock verdict is sustained nominal generated duration divided by producer wall time, combined with the delivery counters and ring trajectory. We still report the normalized quantiles because they decompose thermal changes and preserve comparison with the earlier campaign.

The final corrected A17 run lasts 610 seconds after audio start, captures 600 seconds of PCM, logs every producer iteration, and records token frames after warmup. A separate five-process campaign provides run-level medians and IQRs for temporal ANE and CPU+GPU requested policies. A matched 610-second CPU+GPU-policy run exposes duration effects.

4.3 Crossover design

For seeds 20260718, 271828, and 1618033, both the MLX sampler and an exact Python implementation of the shipped Core ML graph/host boundary generate 15,001 token frames, enough for 600 seconds after decoder lookahead. Each uses its native seeded random-number implementation; token-source arms are therefore independent trajectories, not token-by-token parity trials. Within every decoder comparison, however, the token file is fixed and hash-identical.

The principal 2x2 crosses token source (MLX or Core ML port) with decoder path (stateful streaming MLX or stateless Core ML plus C++ DSP). A fifth arm replaces only the Core ML decoder graph with the MLX FLOAT32 pre-iSTFT graph while preserving 25-token windows and C++ reconstruction. Four more arms cross both token sources and decoder graphs with the corrected periodic-Hann DSP. The causal intervention then repeats MLX and Core ML graphs with 12 frames of left context and identical corrected DSP.

For each 30-second window we measure the fraction of RMS-envelope spectral power from 4 to 16 Hz. The original 0.070 limit was frozen for one ambient prompt before this study. We retain it as a diagnostic count, not a universal music-quality boundary: legitimate rhythm occupies the same band. Primary inference uses paired effect sizes over all windows and reports the median and range of seed-level means. We do not treat 60 temporally adjacent windows as 60 independent samples.

4.4 Independent evidence and receipts

The decoder-context tensor probe compares a stateful MLX prefix against stateless 25-token evaluations at one fixed segment while varying retained history over 0, 1, 2, 4, 8, and 12 tokens. The production C++ reconstruction has an independent sample-level parity test against a NumPy periodic-Hann inverse STFT and overlap-add implementation.

Every public result is generated from a machine-readable receipt. Raw device WAVs, events, tokens, and traces remain private because the product runtime is private, but their SHA-256 hashes, normalized summaries, commands, model identities, and source hashes are public. Precision experiments stop in the order finite, deterministic parity, device latency, then audio; a wrong function never earns a favorable speed number.

5 Results

Table 2: Sustained foreground results. Cost is a p99 over 25-token iteration means, not per-token tail latency. The corrected A17 runtime sustains throughput and playback; its 21 s queue does not satisfy low-latency steering. The CPU+GPU control and A14 rows predate the decoder-context repair and are conservative duration/hardware controls.

Device	p50	p90	p99	Rate	First underflow	Verdict
	iteration-normalized ms/token			(×RT)		
A17 ANE + context	23.50	24.33	24.81	1.030	none / 610 s	throughput pass
A17 CPU+GPU	23.56	43.55	49.23	1.008	none / 610 s	compute fail
A14 ANE	43.88	47.25	58.59	0.897	294.08 s	fail

5.1 Reliable ANE admission and GPU absence

Both devices pass 10/10 cold-process temporal admission and state gates. The A17 trace contains 653 temporal ANE predictions and 24 decoder ANE predictions, plus expected CPU depth activity. The application's Metal GPU table contains zero intervals and zero duration. The A14 cross-device trace reaches the same placement verdict. Moving recurrence to ordinary tensor I/O therefore makes admission repeatable in the application without deleting transformer math.

5.2 Sustained A17 throughput, with the metric named honestly

The corrected A17 context-12 runtime completes 628 nominal one-second producer iterations over 609.65 seconds of producer-loop wall time, a 1.030x nominal rate. It reaches `serious` thermal state and records zero callback underruns, zero producer drops, and no near-zero dropout longer than one sample. The PCM capture is exactly 600.0 seconds, finite, unclipped, and stereo. table 2 summarizes the physical-device results.

Iteration-normalized stage cost is 23.50 ms p50, 24.33 ms p90, and 24.81 ms p99. Decoder context roughly doubles decoder cadence: decoder work rises to 58.74 ms per 25-token iteration at p50, while temporal and depth remain 280.01 and 246.14 ms. Backpressure occupies 38.9% of producer-loop wall time, direct evidence that sustained capacity exceeds playback.

The delivery statement is narrower. Queue fill starts at 2.88 seconds, never falls below 2.73 seconds, and ends at 21.01 seconds, near the configured high watermark. That proves safe continuous playback for the measured run. It also quantifies why the default runtime is not a low-latency steering system.

5.3 Duration changes the placement verdict

Table 3: Independent-process pre-context latency campaign, five runs per cell. Entries are median (IQR) across run-level percentiles. The CPU+GPU control changes only the requested temporal policy; depth remains CPU and decoder remains ANE.

Device	Temporal policy	p50	p99	Temporal p50	Startup s
		iteration-normalized ms/token			
A17 Pro	ANE	20.77 (.03)	22.35 (.42)	11.25 (.01)	4.26 (.04)
A17 Pro	CPU+GPU	27.32 (.25)	38.51 (1.78)	16.35 (.15)	6.22 (.05)
A14	ANE	40.38 (1.00)	50.72 (3.30)	13.85 (.63)	7.52 (.01)
A14	CPU+GPU	49.73 (1.11)	50.64 (1.41)	24.85 (.17)	9.76 (.14)

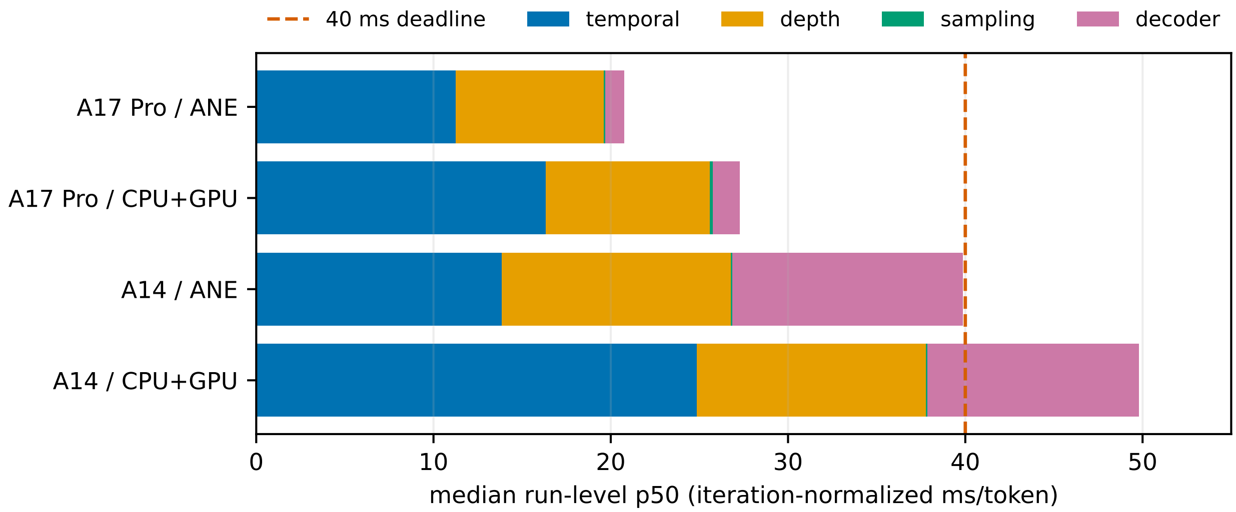

Figure 2: Stage decomposition for the repeated-process campaign. Bars use the median of run-level stage p50 values. The selected ANE policy reduces temporal cost on both phones. These are quantiles of iteration means, not per-token latency quantiles. CPU+GPU denotes a requested Core ML policy, not a GPU-placement proof.

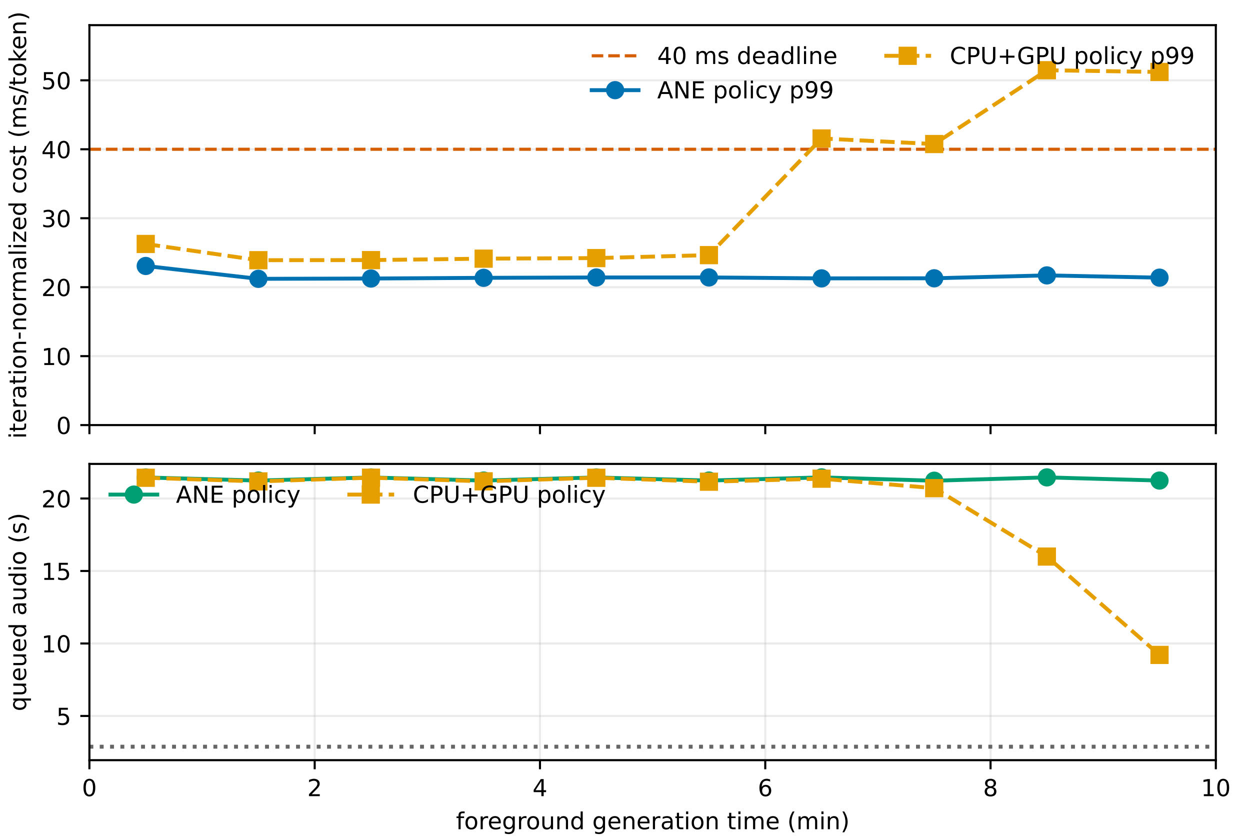

Figure 3: A17 Pro matched pre-context ten-minute trajectories. The selected ANE-policy run keeps iteration-normalized p99 flat. The CPU+GPU-policy control moves from 38.51 ms in five short processes to 49.23 ms here and crosses 40 ms after minute six. The duration effect is visible even though buffering prevents an underrun.

The five-process pre-context campaign reports A17 median run-level p99 iteration-normalized cost of 22.35 ms for temporal ANE and 38.51 ms for the CPU+GPU requested-policy control (table 3, fig. 2). In the matched 610-second control, the latter becomes 49.23 ms. Its per-minute p99 crosses 40 ms only after minute six and reaches 51.46 ms, while a banked reservoir drains from about 21 seconds to 7.65 seconds (fig. 3). A five-minute benchmark would have selected a policy that fails the ten-minute capacity criterion. The selected ANE trajectory begins hotter yet remains flat.

These pre-context controls do not estimate the exact cost of the repaired decoder cadence; the final row above does. They isolate a separate temporal placement and duration effect using identical historical decoder work.

5.4 Crossover localizes the audio defect

Table 4: Three-seed, 600 s crossover. Effects are changes in 4–16 Hz envelope pulse share; entries are median seed mean [range]. The context intervention is paired against the same graph and corrected DSP without context.

Intervention	Effect	Positive windows
Token source at MLX decoder	+.00181 [-.00057, .00627]	35/60
Core ML vs MLX decoder graph	+.00018 [.00016, .00088]	44/60
Stateless window vs streaming MLX	+.01706 [.01613, .01796]	60/60
Add 12-frame context, same Core ML/DSP	-.01650 [-.01772, -.01574]	0/60

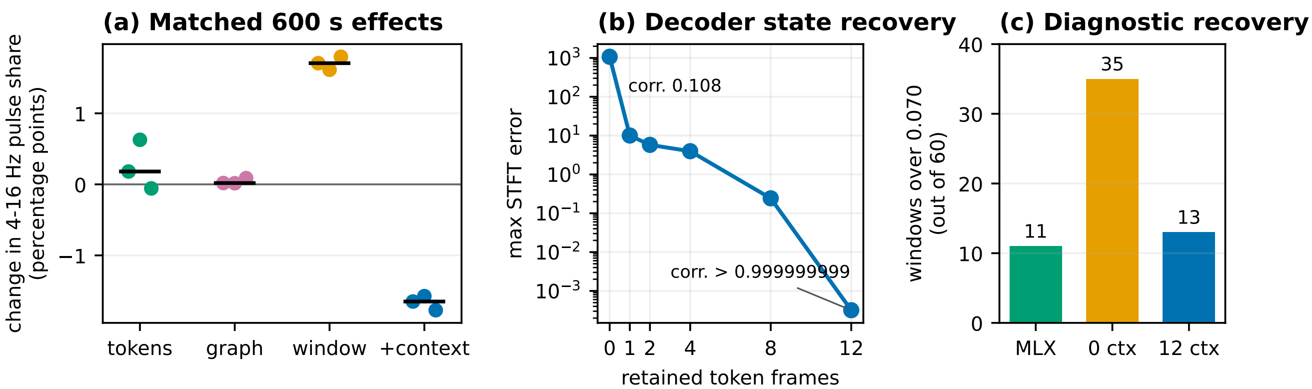

Figure 4: Causal localization. (a) Each dot is one 600 s seed; black bars are medians. (b) Twelve retained token frames recover stateful MLX pre-iSTFT output to correlation above 0.999999999. (c) The original prompt-specific 0.070 diagnostic fires in 35/60 stateless Core ML windows, but only 13/60 corrected windows versus 11/60 stateful MLX windows.

The crossover result is stable across all three seeds (table 4, fig. 4). Replacing MLX tokens with Core ML-port tokens at the stateful MLX decoder has median seed effect +0.00181, with range -0.00057 to +0.00627. Replacing the FLOAT32 MLX decoder graph with the FP16 Core ML graph under identical stateless windows and DSP has effect +0.00018 [0.00016, 0.00088]. Neither is large enough to explain the symptom.

Stateless windowing and the legacy production DSP, by contrast, add 0.01706 [0.01613, 0.01796] to pulse share relative to streaming MLX, and the paired difference is positive in all 60 windows. The FLOAT32 MLX graph reproduces the effect, exonerating FP16 as its primary source. Correcting the Hann convention without context adds another 0.00321 rather than removing it.

The context probe identifies the missing state. With no left history, a stateless 25-token evaluation has correlation 0.1083 with the corresponding stateful pre-iSTFT tensor and maximum absolute error 1,071.7. Correlation is 0.9926 with one token, 0.9999955 with eight, and above 0.999999999 with 12; maximum error falls to 0.000319. Under the same graph and corrected DSP, adding 12-token context reduces audio pulse share by 0.01650 [-0.01772, -0.01574], negative in all 60 paired windows.

The original 0.070 diagnostic fires in 35/60 stateless Core ML windows, 13/60 context-corrected windows, and 11/60 stateful MLX windows. On the physical A17 capture, median window share is 0.06091 and 5/20 windows cross the limit - exactly the stateful MLX count for seed 20260718. The final window is 0.09181, which reinforces why a single tail threshold is not a universal clock. The full trajectory contains no short exact token cycle, no sustained dropout, and no monotone decoder-specific late-onset failure.

The evidence therefore rejects the paper's earlier simple claim of runtime-independent long-horizon recurrence. It does not prove that MRT2 can never degenerate, or that every prompt is high quality. It proves that the measured pulse excess attributed to the phone path is caused primarily by an incorrect causal decoder boundary and is repaired by restoring measured history.

5.5 A14 is below the sustained boundary

The A14 story remains a negative lower bound. Before the context repair, its certified FLOAT32 decoder configuration measured 43.88, 47.25, and 58.59 ms iteration-normalized p50/p90/p99, produced at 0.897x, and exhausted a 20.16 second reservoir after 294.08 seconds. It recorded 7,952 underruns. Context repair increases decoder cadence, so it cannot rescue this capacity result.

The A17 FP16 decoder artifact is not promoted on A14. A short ANE probe on A14 is finite but has unsafe amplitude; a long retry later becomes non-finite, while FLOAT32 CPU isolation remains bounded. We did not localize the exact overflowing layer, so we report a device-artifact numerical incompatibility, not a layer-level mechanism. A14 needs a different trained precision or decoder/depth architecture, not a larger hidden buffer.

5.6 Compression ladder and early stopping

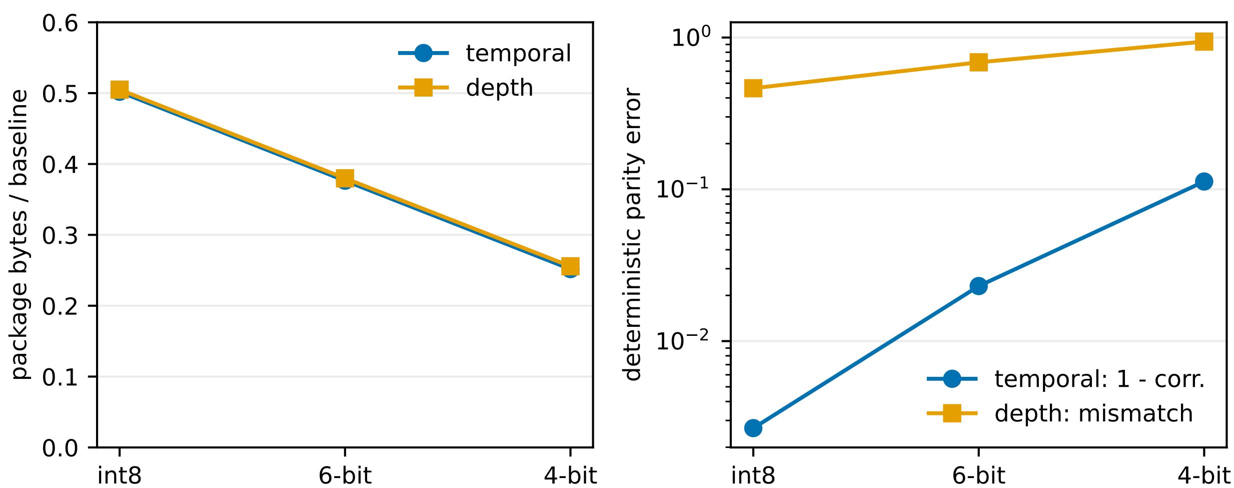

Figure 5: Rejected post-training compression ladder. Every artifact remains finite and becomes substantially smaller, but deterministic parity degrades monotonically. The gate order stops all six candidates before device timing or listening, avoiding a speed result for the wrong function.

We also test int8 linear quantization and 6-bit and 4-bit palettization on temporal and depth packages (fig. 5). Temporal packages shrink to 50.2%, 37.6%, and 25.2% of baseline, but 64-step correlation falls to 0.9973, 0.9770, and 0.8874, below the 0.999 gate. Depth argmax mismatch reaches 46.3%, 68.7%, and 94.0%. All remain finite; none reaches device timing. This is a negative result about tested post-training methods, not a claim that trained low-precision MRT2 is impossible.

6 Discussion

6.1 Why the crossover changes the science

The first manuscript had disciplined receipts and the wrong causal unit. It compared a 600-second phone capture with short clean MLX excerpts, then named a failed generative clock. The crossover forces every explanation to predict a path through fixed tokens. Because the FLOAT32 MLX pre-iSTFT graph reproduces the phone effect and

12-frame context cancels it under both graphs, the result is stronger than either "the model degenerates" or "Core ML is numerically different." It is a specific state-contract error with a minimal intervention.

This pattern generalizes. Causal neural decoders may expose a formal lookahead that is not their complete left-receptive-field contract. Exporting a static window without explicit state turns missing history into deterministic boundary modulation. Output finiteness, tensor shape parity, seam-jump metrics, and even short perceptual clips can all pass. A useful export test must compare a stateless window with a stateful prefix at multiple history depths.

6.2 Throughput is not frame latency

Dividing a batch time by 25 produces a unit of ms/token, not 25 observations. Its quantiles describe iteration-level compute density. They average away within-batch stalls and should not be labeled p99 frame latency. Our corrected compute clock uses the variable the architecture actually needs: sustained audio production relative to playback. Per-token tail latency would require instrumenting individual temporal/depth steps, which the current production log does not do.

This correction does not weaken the thermal result. The reservoir trajectory, producer rate, and late CPU+GPU-policy slowdown are batch-appropriate evidence. It does narrow the headline: we demonstrate sustained generation throughput, not a frame-synchronous 40 ms scheduler.

6.3 Continuous playback is not interactive liveness

A growing or high-watermark-limited reservoir prevents underflow and delays intent. The old delivery criterion accepted any final queue larger than the initial queue. For a continuously steerable model, that condition rewards the wrong behavior. A defensible upper bound is derived from prompt-change-to- audible-change latency, not memory capacity.

The discard experiment offers an architectural direction: retain 0.4 seconds, apply a short fade, and count intentional discard separately from producer drops. But it has not passed a human speaker-level click-free test. The current system is therefore a robust buffered playback engine and a candidate live engine. Calling both simply "real time" would erase the most important product difference.

6.4 Evidence paths are systems too

The project exposed two evidence-path bugs. An early recorder flattened a padded FP16 `MLMultiArray`, manufacturing huge coefficients that live playback never consumed. The decoder's reconstruction core then used a different Hann convention from upstream. The first bug invalidated a capture; the second was a real parity debt but not the observed root cause. Both are now covered by independent stride-aware and sample-level tests.

The methodological rule is simple: test the measuring instrument with known good and known bad controls, then cross implementations before assigning a failure to the model. Hashes and manifests preserve evidence; they cannot make an underdetermined experiment causal.

7 Limitations

The crossover uses three seeds and one ambient prompt. That is enough to replicate the decoder-boundary effect, not to characterize the prompt distribution or long-horizon musical quality of MRT2. The 4-16 Hz measure is prompt-sensitive and overlaps legitimate musical rhythm; its 0.070 value is reported only as the original diagnostic boundary. We do not use the prior five automated listening votes as independent statistical evidence: two late excerpts overlap, and an audio model is not a substitute for counterbalanced human listening.

The MLX and Core ML token generators use different native random-number schemes. Token-source arms therefore test distributions and downstream symptoms, not token-exact port equivalence. Temporal fixtures and FLOAT32 depth rollout provide separate deterministic parity evidence, but rare FP16 near-tie flips remain expected. The context tensor probe uses one segment; three-seed full-audio interventions provide the broader replication.

The final corrected runtime has one 610-second physical-device run. The placement campaign, pre-context latency campaign, and A14 study reuse prior signed builds with identical model artifacts; the context repair is host-only. The A14 FP16 failure is not localized to a layer. We report Instruments impact and placement evidence but no calibrated joules or battery-life claim.

Finally, the default 21-second queue is incompatible with a strong live- steering claim. The 0.4-second discard path needs counterbalanced human speaker-level evaluation. The complete Crossfade runtime and raw audio remain private; public artifacts contain exporters, validators, normalized receipts, hashes, figure sources, and manuscript source. The runtime is available from the author for artifact review.

8 Conclusion

MRT2 can sustain GPU-free generative-audio throughput on an A17 Pro iPhone under a ten-minute foreground thermal load. The corrected system produces at 1.030x nominal real time with zero underruns or drops, while a pure temporal boundary makes ANE admission repeatable. Those are substantial systems results, but they are not synonymous with per-token tail latency or low-latency steering.

The strongest result came from trying to falsify our own negative claim. A three-seed token-by-decoder crossover showed that an apparent long-horizon generative failure followed stateless decoder windowing, not the model, token source, FP16 graph, or Hann reconstruction. Twelve frames of measured causal history recover tensor parity and remove the excess on Mac and iPhone. Live generation has three clocks. Good science requires measuring each one and, when a clock fails, crossing the boundary until the cause has nowhere left to hide.

9 Artifact statement

The public repository contains Core ML exporters, frozen fixtures, placement and sustain verifiers, the three 600-second crossover reports, decoder-context probe, corrected A17 normalized manifest, compression ladder, figure sources, paper source, and exact build protocol. Public reports hash private WAVs, tokens, events, traces, signed executables, and relevant runtime sources. Preconverted artifacts are mirrored on Hugging Face. The product application and raw listening media are available from the author for artifact review.

10 Acknowledgments

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A Reproducibility

A.1 Public evidence map

The canonical crossover result is `aggregate.json` in the public crossover receipt directory. It hashes three seed reports and the decoder-context probe. Each seed report hashes token summaries and every decoded WAV arm. The corrected physical-device receipt is `context12-soak-manifest.json` in the A17 context-12 directory; it hashes the 600-second PCM capture, 610-second event trace, 15,790-frame token capture, normalized summaries, signed executable, runtime source, and render core source.

The A17 and A14 placement directories contain admission receipts. The historical A17 placement soak and CPU+GPU duration control are under `a17pro/soak/` and `evaluation/`. The A14 bounded-reservoir result is under `a14/soak/`. `MRT2WeightCompressionLadder.json` contains all six early-stopped compression arms, and `depth-rollout-ablation.json` records the measured 12-call, one-call FLOAT32, and one-call FLOAT16 depth results.

A.2 Crossover commands

The private Crossfade harness exposes generation, decoding, summarization, and decoder-context probe subcommands. A complete replication uses 15,001 codebook-local token frames, temperature 1.0, top-k 40, a 10-second refresh, and a 600-second decode. Decoder arms name the token file explicitly; the analyzer refuses incomplete fixed-DSP or context pairs.

The public analysis commands are:

- run `analyze_system_paper_crossover.py` once per seed with the four 2x2 WAVs, FLOAT32 graph split, four corrected-DSP controls, and two context arms;
- run `aggregate_system_paper_crossover.py` on the three reports plus `decoder-context-probe.json`; and
- regenerate `fig-crossover.pdf` from the aggregate only.

A.3 Corrected runtime protocol

1. Load the signed precompiled models with explicit compute policies, query the temporal compute plan, execute the fresh/warmed state proof, and perform three untimed warmups.
2. Prime the PCM ring. For each token frame, present chronological temporal K/V, run one temporal prediction, sample 12 depth levels in one prediction, and sum selected RVQ rows.
3. Once 25 decoder embeddings are available, predict 96 STFT frames. After the first window, retain 12 token embeddings, advance by 12, discard 48 context STFT frames, and send only 48 new frames to the periodic-Hann inverse STFT.
4. Push PCM off the callback. The callback performs bounded ring reads; count zero-fill underruns. Backpressure at the high watermark. Every 10 seconds, clear temporal K/V and previous feedback without clearing decoder context, overlap-add state, or queued PCM.
5. Capture logical STFT-derived PCM and codebook-local tokens independently; stop after 610 seconds and hash every artifact before normalization.

A.4 Verification suites

The public suites cover crossover audio analysis, seed aggregation, G1-G4 manifests, placement, sustain, and compression contracts. The private runtime suites cover Swift configuration defaults, padded `MLMultiArray` reads, exact SplitMix64 Gumbel values, cache updates, token-cycle metrics, C++ periodic-Hann sample parity, and the signed iOS build. Tectonic rebuilds the paper from the readable Markdown source, and Poppler-rendered pages are visually audited before publication.

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